

# Kidney Allocation Under Observational Learning: A Comparison of Four Mechanisms

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## 1 The Real-World Problem

When someone dies and becomes an organ donor, their kidneys must be transplanted quickly. A kidney removed from a deceased donor stays viable for transplant for roughly 12 to 24 hours before the tissue degrades enough that doctors will no longer use it. After that, the kidney is discarded—wasted—even though tens of thousands of patients are waiting for one. The U.S. discards roughly 22% of procured deceased-donor kidneys, a number that has been rising rather than falling over the past decade (Mohan et al., 2023). The French system, with similar donor characteristics, discards under 10%.

The current U.S. allocation system works roughly like this. When a kidney becomes available, the procurement organization consults a ranked list of waiting patients. The patient at the top of the list is offered the kidney first. Their transplant center has up to one hour to say yes or no. If they say no, the offer moves to the next patient. And so on, down the list. Each “no” eats one hour off the kidney’s viability clock. If twelve people in a row decline, the kidney has used its entire window and gets thrown away—even though the thirteenth person, or the twentieth, or the hundredth, might have said yes.

This is one of the cleanest examples in market design of a fundamental tension: **the mechanism that matches goods to people has a budget**, and the budget gets eaten by declines. Husain et al. (2019) report that the median recipient of a deceased-donor kidney received 17 prior offers (made on behalf of higher-priority patients) before the kidney came to them. 30% of waitlisted patients die or are removed from the list before ever receiving a transplant. The kidney sitting at the top of the search list cannot wait for the right answer—it has to find an answer in 12 hours.

This study compares four mechanisms for solving that problem. We build a single model of the world and patient behavior, and then simulate each mechanism inside that world. The question is which mechanism produces the most transplants, with the right matches, without unfair treatment of patients waiting in line.

## 2 The Four Mechanisms

Before going into the model, let’s name the four mechanisms we’ll compare. Each one is a different policy choice about how to find a recipient for an available kidney.

**Omniscient.** A first-best benchmark. The mechanism magically knows every patient’s preferences and instantly identifies the highest-priority patient who would accept this kidney. It then gives them the kidney with no further friction. This is unattainable in practice—real allocators don’t see preferences and can’t act instantly—but it tells us the upper bound for what any mechanism could achieve.

**Sequential.** The status-quo mechanism. The mechanism offers the kidney to one patient per hour, walking down the priority queue, until someone accepts or 12 hours elapse. This is approximately how OPTN allocation works today, abstracting away from some complications (parallel offering to multiple centers, geographic priority bands, etc.).

**Batching.** A proposed reform. Instead of offering to one patient per hour, the mechanism offers simultaneously to the next  $m$  patients in priority order. Patients have one hour to respond. Whoever among them says yes—preferring the highest-ranked acceptor if multiple do—gets the kidney. If all  $m$  decline, the next batch is offered the next hour. With  $m = 5$  and 12 hours, the mechanism can reach up to 60 candidates rather than 12. The Organ Donation and Transplantation Alliance has been piloting batched offers, finding that batches of four reduce workload and improve utilization.

**Threshold.** A more aggressive redesign, based on Su and Zenios (2005). Each patient pre-declares which kidney quality ranges they would accept. When a kidney arrives, the mechanism identifies patients who have declared this quality range acceptable and offers to them in priority order. Non-declarers are skipped entirely. We test several versions of this mechanism corresponding to different ways of asking patients about their preferences—a single binary opt-in (which is what OPTN currently uses for high-KDPI kidneys), three buckets, five, ten, and a continuous self-declaration.

To compare these mechanisms fairly, we need every patient to behave the same way regardless of which mechanism they face. The mechanism doesn’t change what the patient wants—it changes how the system finds out what the patient wants and how often it asks. So we first specify a single model of patient behavior, then run all four mechanisms against that same world.

## 3 The Model of the World

### 3.1 Kidneys

A kidney has a quality  $\theta \in [0, 1]$ , where  $\theta = 1$  is the best possible kidney and  $\theta = 0$  is the worst. This  $\theta$  maps directly to the OPTN Kidney Donor Profile Index (KDPI): a kidney with KDPI = 85 (a “marginal” kidney that the OPTN flags as requiring special consent) has  $\theta = 0.15$ . A premium kidney with KDPI = 15 has  $\theta = 0.85$ .

Because KDPI is a percentile score by construction, the distribution of  $\theta$  across the donor population is uniform on  $[0, 1]$ . That’s a definitional consequence of using percentiles—half of donors have KDPI below 50, by definition. So we draw kidney qualities as  $\theta \sim \text{Uniform}[0, 1]$ .

Kidneys arrive one at a time, indexed  $k = 1, 2, 3, \dots$ . Each kidney has 12 hours of viability before discard.

### 3.2 Patients

Each patient  $i$  has a **reservation threshold**  $\tilde{\theta}_i \in [0, 1]$ . They are willing to accept a kidney if and only if its quality is at least their threshold:

$$\text{Patient } i \text{ would accept kidney } \theta \iff \tilde{\theta}_i \leq \theta.$$

This is the patient’s *preference*, distinct from their *behavior* (which we model in the next subsection). A patient with  $\tilde{\theta}_i = 0.05$  has very weak outside options and will accept nearly any kidney. A patient with  $\tilde{\theta}_i = 0.50$  is more selective and will only accept moderate or better kidneys. A patient with  $\tilde{\theta}_i = 0.95$  is extremely selective.

We draw thresholds from a  $\text{Beta}(2, 5)$  distribution, which has mean  $\approx 0.29$  and is right-skewed. This corresponds to a population where most patients are willing to accept moderate kidneys but only a minority would accept the most marginal ones. The calibration matches roughly 30% of OPTN waitlist patients who opt in to receive high-KDPI kidneys.

The threshold  $\tilde{\theta}_i$  is **private information**—only the patient knows it. The mechanism cannot read minds. Different mechanisms infer thresholds differently: omniscient sees them magically, sequential never asks about them, batching never asks, and threshold mechanisms ask patients to declare them ahead of time.

Patients arrive at rate  $\rho = 1.6$  per kidney, matching the real U.S. ratio (roughly 40,000 new waitlist entries per year against 25,000 deceased-donor transplants). The queue grows over time. Priority is determined by arrival order—earlier arrivals get higher priority. This abstracts from the actual OPTN priority system, which uses time-on-list, medical urgency, prior living donor status, and several other factors. We’re treating priority as if it were purely time-on-list.

### 3.3 The Behavioral Model: How Patients Actually Respond to Offers

Suppose patient  $i$  is offered kidney  $\theta$ . They are at queue position  $n$  when offered, and they have observed  $k$  prior declines for this kidney (because they’re the  $(k + 1)$ -th person to be asked). What is the probability they accept?

We use a two-stage rule. **First, the threshold gate:** if  $\tilde{\theta}_i > \theta$ , the patient declines for certain. They literally would not want this kidney, regardless of position or what others have done. **Second, the logistic:** among the offers that pass the threshold gate, acceptance is noisy. Patients sometimes decline for logistical reasons even when the kidney is acceptable.

Formally,

$$P(\text{accept} \mid \text{position } n, k \text{ prior declines}) = \begin{cases} 0 & \text{if } \tilde{\theta}_i > \theta \\ \sigma(\alpha + \beta(n - 1) - \delta k) & \text{if } \tilde{\theta}_i \leq \theta \end{cases}$$

where  $\sigma(x) = 1/(1 + e^{-x})$  is the logistic function.

The three parameters mean the following:

- $\alpha = \text{logit}(0.9) \approx 2.197$ . This sets the baseline acceptance among threshold-eligible patients to 90% (when position is 1 and no priors have been observed). The 10% baseline decline captures non-preference reasons: surgical scheduling conflicts, last-minute donor concerns, family decisions.
- $\beta = 0.10$ . This is the **position effect**. Patients deeper in the queue accept more readily, even conditional on threshold being met. The interpretation is desperation: someone at position 30 has been waiting longer, has been declined for many prior kidneys, and is more willing to overlook logistical inconveniences. Each step deeper increases the log-odds of acceptance by 0.10, so position 30 accepts with  $\sigma(\alpha + 2.9 - \delta k)$ —substantially higher than position 1.
- $\delta$  is the **lemons strength**. Each prior decline observed for this kidney decreases the next patient’s log-odds of acceptance by  $\delta$ . We sweep over  $\delta \in \{0, 0.3, 1.0\}$ . At  $\delta = 0$ , prior declines have no effect. At  $\delta = 1.0$ , each prior decline cuts acceptance odds by a factor of  $e \approx 2.7$ .

The lemons term is the crucial behavioral feature of this study. It is named after Akerlof (1970)’s “The Market for Lemons” and captures the empirically documented fact that, in real OPTN data, candidates lower on the priority list refuse marginal kidneys more often than expected, even after controlling for everything observable about the kidney (Zhang, 2010). The interpretation is

**observational learning:** when previous patients in line decline a kidney, the next candidate quite reasonably starts to wonder if there’s something wrong with it that the medical paperwork doesn’t capture. Maybe there’s an issue the earlier centers spotted. Maybe the donor had a history that’s only obvious to careful readers. Each rejection makes the next candidate more cautious.

A key subtle point is that **the lemons signal doesn’t distinguish between “they didn’t want a kidney this marginal” and “they declined for medical/logistical reasons.”** All a downstream patient sees is that the previous patient said no. They don’t know why. So both kinds of decline contribute to the lemons signal: the auto-declines (threshold violations) and the noisy declines (logistic failures). This matches Zhang’s empirical setup.

### 3.4 Why This Behavioral Model

A few choices need defending. First, why a threshold gate? In a frictionless world with no information cost, patients would always accept any kidney above their reservation utility and decline any below. The threshold gate captures this. Without it, the model would predict patients accepting kidneys they would later regret accepting—and a lot of empirical work suggests this regret should be rare in equilibrium.

Second, why an additional logistic on top? A pure threshold model would predict 100% acceptance among eligible patients, which is empirically false. Real-world OPTN data shows decline rates well above zero even among patients who pre-opted-in to receive marginal kidneys. The 10% baseline decline captures the universe of things that can go wrong with a transplant offer that aren’t about the patient’s preferences. This is consistent with the Husain et al. (2019) finding that even high-acceptance patients show decline rates of 10–20%.

Third, why a position effect? In reality, queue depth correlates with several things—waiting time, medical urgency, recent failed offers from competing kidneys—that should push acceptance up. A patient at the front of the queue has the option to wait for a better kidney. A patient deep in the queue does not. Empirically, deeper-positioned patients show higher acceptance rates than shallower ones on observably identical kidneys. The logistic position term  $\beta(n - 1)$  is a simple way to put this in the model.

Fourth, why a lemons effect? This is the contested feature in the literature. Zhang (2010) provides direct empirical evidence for observational learning in U.S. kidney offer responses. Agarwal et al. (2021) propose an alternative interpretation in which declines reflect privately observed kidney characteristics (so deeper patients aren’t *learning* from prior declines—they’re seeing the same kidney problems independently). The two interpretations predict similar static behavior but very different responses to mechanism changes. Our model takes Zhang’s interpretation at face value because it gives mechanisms the strongest “lever” to act on, making the comparison sharper. Our results should be read as conditional on Zhang being right.

This is a single behavioral model, applied identically across all four mechanisms. **What changes across mechanisms is which patients get offered the kidney, when, and in what configuration. The patient’s response to a given offer is the same in every mechanism.**

## 4 The Mechanisms in Detail

We now describe how each mechanism takes the world and produces an allocation outcome for each kidney.

## 4.1 Omniscient

Omniscient is straightforward. When kidney  $\theta$  arrives, the mechanism scans the queue from front to back and gives the kidney to the first patient with  $\tilde{\theta}_i \leq \theta$ . If no such patient exists, the kidney is discarded.

This mechanism deliberately ignores the behavioral model—there’s no logistic, no lemons. It assumes the mechanism can directly observe preferences. It also has unlimited search depth: it will walk to position 1000 if needed.

Omniscient is the first-best benchmark. We don’t propose it as a real policy. We use it to measure the gap between what we achieve and what we *could* achieve in principle. Any discard rate above omniscient’s is a “loss” attributable to information frictions: limited search budget, decline noise, or lemons cascades.

## 4.2 Sequential

When kidney  $\theta$  arrives, the mechanism starts at position 1 in the queue. Patient 1 is offered the kidney, with acceptance probability following the behavioral model above. If they accept, they get the kidney. If they decline (either via threshold gate or logistic noise), the mechanism moves to patient 2 in the next hour. The kidney expires after 12 hours of offers—so sequential can reach at most 12 patients.

Sequential is the closest approximation to current OPTN practice. It is the mechanism the audience is most familiar with intuitively. It also exhibits the lemons cascade in its purest form: each decline at position  $n$  pushes the lemons signal  $k$  from  $n - 1$  to  $n$ , lowering acceptance at the next position.

## 4.3 Batching

Same as sequential, except instead of offering to one patient per hour, the mechanism offers simultaneously to  $m$  patients per hour. We use  $m = 5$  throughout, based on Part 1’s finding of diminishing returns past  $m = 3$  and OPTN pilots that use batches of 4–5.

The crucial feature of batching is that **patients within a batch act on the same information**. All five patients see the same  $k$  at the time of their decision. They don’t see each other’s responses. This freezes the lemons signal within batches: the cascade only updates at batch boundaries, not at every position.

Concretely, in a batch of 5, the kidney’s lemons signal updates as  $k = 0, 5, 10, \dots$  rather than  $0, 1, 2, \dots$ . With 12 hours and  $m = 5$ , batching reaches up to 60 candidates (5 in each of 12 batches), versus sequential’s 12 candidates. So batching has two simultaneous effects: it sees more patients per hour, and it preserves better information for those patients.

Among accepters in the same batch, the kidney goes to the highest-priority one. So a patient at position 1 who would have gotten the kidney under sequential still gets it under batching, as long as they accept; batching only changes who gets the kidney if position 1 declines.

## 4.4 Threshold

Threshold mechanisms are based on the choice-based allocation framework of Su and Zenios (2005). The key idea is that patients themselves know things about their preferences that the mechanism doesn’t. So why not just ask?

In a threshold mechanism, each patient declares in advance which range of kidney qualities they would consider acceptable. Specifically, the kidney quality space  $[0, 1]$  is partitioned into  $K$  buckets,

and each patient declares which bucket their reservation threshold falls into. When a kidney of quality  $\theta$  arrives, the mechanism identifies all patients who declared a bucket that includes  $\theta$  or below, and offers to them in priority order. Patients who declared a bucket strictly above the kidney’s quality are skipped—the mechanism knows in advance they would decline.

The patient’s *declaration* is one of  $K$  options. Their *true* threshold is a real number. So even within a bucket, some patients might decline a particular kidney because the kidney falls below their precise threshold (even though it’s above the bucket boundary). This residual misalignment generates within-bucket declines, which the lemons signal then picks up.

For an example: suppose the cutoff is at  $\theta = 0.15$  (this is the current OPTN policy for high-KDPI kidneys). A patient with reservation threshold  $\tilde{\theta}_i = 0.05$  declares “yes” to bucket 0 (which contains  $[0, 0.15)$ ). A patient with  $\tilde{\theta}_i = 0.13$  also declares “yes” to bucket 0—their true threshold is still within bucket 0’s range. When a kidney of quality  $\theta = 0.10$  arrives, both patients are eligible for offers. Patient #1 would accept ( $0.10 > 0.05$ ). Patient #2 would decline ( $0.10 < 0.13$ ). So even when patients honestly declare, coarse buckets can produce within-bucket mismatches.

We test five versions of the threshold mechanism, varying  $K$  and the cutoff structure:

- **K=2 (OPTN)**: a single binary cut at  $\theta = 0.15$  (corresponding to KDPI = 85). Patients opt in or out of receiving “high-KDPI” kidneys. This is the current OPTN policy.
- **K=3**: three uniform buckets at cuts  $\theta = 1/3, 2/3$ .
- **K=5**: five uniform buckets at cuts  $\theta = 0.2, 0.4, 0.6, 0.8$ .
- **K=10**: ten uniform buckets.
- **K= $\infty$  (continuous)**: patients declare a continuous threshold. The mechanism uses the patient’s exact reservation threshold for the eligibility check. This is the Su–Zenios theoretical optimum.

Within the eligible set, the mechanism still proceeds sequentially (one offer per hour). The behavioral model applies as before. So threshold mechanisms eliminate the kind of decline that comes from “the kidney is below my threshold,” but they still suffer from the logistic noise and from lemons effects when within-bucket mismatches generate declines.

The 12-hour budget under threshold mechanisms applies to the number of *eligible* patients offered—not to the total number of patients in the queue. Skipping an ineligible patient costs the mechanism nothing in time. This is the key efficiency gain of threshold mechanisms.

A note on incentive compatibility. Under Su and Zenios (2005), if patients know their own reservation thresholds and outside options, declaring truthfully is a dominant strategy. They have no reason to misrepresent their preferences—declaring a more selective bucket means losing offers they would have accepted; declaring a less selective bucket means accepting kidneys they would have preferred to decline. We assume patients declare truthfully throughout. This is a strong assumption that the structural literature (Agarwal et al., 2025) does substantially more careful work to justify in real data.

## 5 Implementing the Simulation

We simulate a stream of  $N = 1500$  kidneys arriving sequentially. The initial queue contains 150 patients drawn from the threshold distribution. After each kidney,  $\rho = 1.6$  new patients on average arrive. The first 300 kidneys are warm-up—discarded from analysis so that the queue reaches a stationary regime. The remaining 1200 kidneys are the “test set” over which we measure outcomes.

We average outcomes over 15 independent replications. Standard errors on discard rates are around 0.001–0.004, so the differences we report below are far larger than Monte Carlo noise.

The mechanisms are run on the same calibrated world, but with mechanism-specific random number streams for the acceptance decisions. For the fairness analysis specifically, we use **common random numbers**: same kidney arrivals, same patient arrivals, same thresholds, same random decision draws—so that we can compare patient-by-patient outcomes across mechanisms in a controlled way. More on this below.

## 6 Headline Results

### 6.1 Discard Rate and Mean Placement, Across All Kidneys

The two primary metrics are discard rate (lower is better) and mean placement among allocated kidneys (a queue-rank measure of how deep the mechanism reached; lower means kidneys went to higher-priority patients on average).

Tables 1, 2, and 3 show the headline numbers across the lemons-strength sweep.

Table 1: Aggregate metrics at  $\delta = 0$  (no observational learning)

| Mechanism                             | Discard rate | Mean placement | Transplant rate |
|---------------------------------------|--------------|----------------|-----------------|
| Omniscient                            | 2.3%         | 22.2           | 47.4%           |
| Sequential                            | 9.4%         | 2.1            | 41.4%           |
| Batching $m = 5$                      | 5.5%         | 4.6            | 44.9%           |
| Threshold $K = 2$ (OPTN)              | 5.6%         | 4.6            | 44.7%           |
| Threshold $K = 5$                     | 5.8%         | 3.7            | 44.5%           |
| Threshold $K = 10$                    | 3.8%         | 7.4            | 46.3%           |
| Threshold continuous ( $K = \infty$ ) | 2.0%         | 21.7           | 47.7%           |

Table 2: Aggregate metrics at  $\delta = 0.3$  (moderate observational learning)

| Mechanism                             | Discard rate | Mean placement | Transplant rate |
|---------------------------------------|--------------|----------------|-----------------|
| Omniscient                            | 2.3%         | 22.2           | 47.4%           |
| Sequential                            | 10.8%        | 2.1            | 40.2%           |
| Batching $m = 5$                      | 8.4%         | 2.6            | 42.1%           |
| Threshold $K = 2$ (OPTN)              | 6.5%         | 4.8            | 44.0%           |
| Threshold $K = 5$                     | 6.0%         | 3.9            | 44.5%           |
| Threshold $K = 10$                    | 3.8%         | 7.8            | 46.1%           |
| Threshold continuous ( $K = \infty$ ) | 2.2%         | 21.7           | 47.5%           |

A few patterns are immediately visible.

**Threshold continuous essentially matches omniscient on discard rate regardless of  $\delta$ .** It loses about 0.3–0.5 percentage points to omniscient because of the 10% baseline decline in the logistic, but the lemons cascade is almost completely neutralized.

**Batching beats sequential at all  $\delta$ ,** consistent with Part 1’s finding. The improvement grows with  $\delta$ : at  $\delta = 0$ , batching saves 4 percentage points; at  $\delta = 1.0$ , it saves 5 percentage points.

Table 3: Aggregate metrics at  $\delta = 1.0$  (strong observational learning)

| Mechanism                             | Discard rate | Mean placement | Transplant rate |
|---------------------------------------|--------------|----------------|-----------------|
| Omniscient                            | 2.3%         | 22.2           | 47.4%           |
| Sequential                            | 19.6%        | 1.5            | 32.7%           |
| Batching $m = 5$                      | 14.4%        | 1.7            | 37.2%           |
| Threshold $K = 2$ (OPTN)              | 15.1%        | 3.8            | 36.6%           |
| Threshold $K = 5$                     | 10.7%        | 2.7            | 40.4%           |
| Threshold $K = 10$                    | 6.3%         | 6.9            | 44.2%           |
| Threshold continuous ( $K = \infty$ ) | 2.6%         | 19.6           | 47.2%           |

**Coarse threshold mechanisms behave non-uniformly.** At low  $\delta$ , the coarse thresholds ( $K = 2$ ,  $K = 3$ ,  $K = 5$ ) perform comparably to batching. As  $\delta$  grows, finer- $K$  thresholds pull ahead while  $K = 2$  does not—at  $\delta = 1.0$ ,  $K = 2$  (15.1%) is essentially tied with batching (14.4%), while  $K = 10$  (6.3%) is much better.

**Mean placement reveals a hidden tradeoff.** Mechanisms that reach deep into the queue (threshold continuous at position 21.7, omniscient at position 22.2) achieve low discards, while mechanisms that stay shallow (sequential at position 2.1) achieve high discards. The threshold mechanism with finer  $K$  trades depth for fewer discards. This raises a fairness concern: are deep-position patients “stealing” kidneys from the front of the queue? We address this directly in §6.5 below.

## 6.2 Pareto Frontier

When we plot discard rate against mean placement, mechanisms form a frontier. The frontier shifts as  $\delta$  changes.

Figure 1 shows the Pareto frontier across the three lemons strengths. At  $\delta = 0$  (left panel), all mechanisms except sequential cluster near the lower-left corner—low discard at low placement. Threshold continuous is far up on the placement axis because it walks deep into the queue, but achieves very low discard. At  $\delta = 0.3$  (middle), sequential pulls away from the cluster and batching deteriorates. The threshold mechanisms maintain their position. At  $\delta = 1.0$  (right), sequential and batching are clearly inferior to all threshold mechanisms except possibly the very coarse  $K=2$  OPTN, which is comparable to batching on both axes.

## 6.3 Discard Rate as a Function of Lemons Strength

Figure 2 makes the central tradeoff visible at a glance. As observational learning strengthens, mechanisms that don’t pre-filter eligible patients deteriorate sharply. Threshold continuous, which only ever offers to true accepters, is essentially immune.

## 6.4 Patient Wait Times

The mechanism affects not just the kidneys (discards) but also the patients waiting in line (waits). For a patient who eventually receives a kidney, the “wait” is the number of kidneys that arrived between their joining the queue and their receiving an organ.

At  $\delta = 1.0$ , the mean wait among patients who received a kidney is:



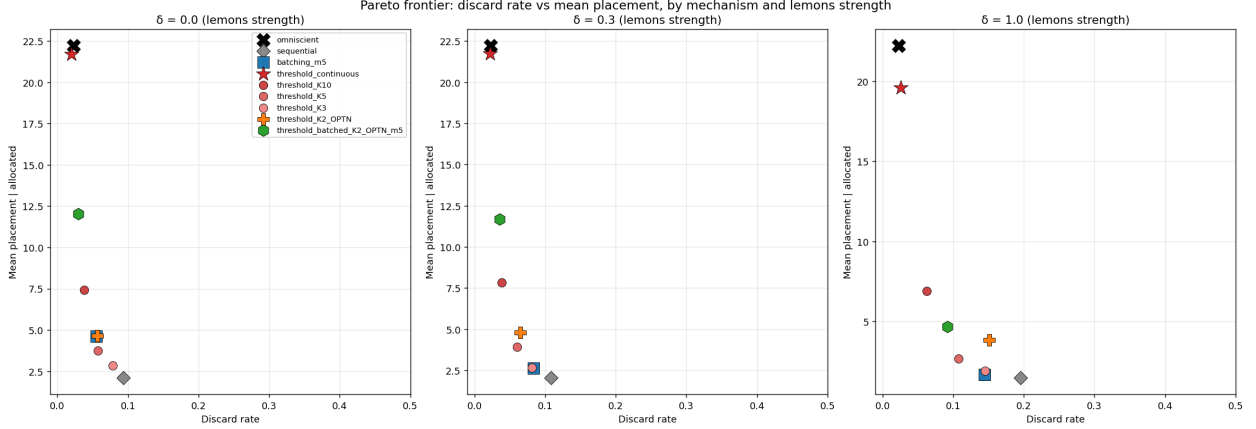


Figure 1: **Pareto frontier of mechanisms in (discard rate, mean placement) space, separately for  $\delta = 0, 0.3, 1.0$ .** Each marker represents one mechanism. Lower-left is better: fewer discards and shallower allocation depth. Mechanisms that fall below and to the left of other mechanisms strictly dominate them on both axes. The omniscient benchmark (black X) sits at high mean placement (around 22) but very low discard rate—it reaches as deep as needed but only to true accepters. Threshold continuous (red star) sits very close to omniscient at every  $\delta$ . Sequential (gray diamond) and batching (blue square) have low mean placement but high discard rate, especially at high  $\delta$ . The coarse threshold mechanisms ( $K = 3, 5, 10$ , pink-to-red circles) interpolate between the extremes. As  $\delta$  grows from 0 to 1.0, the entire frontier shifts rightward (more discards) for mechanisms with no self-selection, while the threshold-continuous mechanism stays put.

| Mechanism                             | Mean wait (kidney count) | Transplant rate |
|---------------------------------------|--------------------------|-----------------|
| Omniscient                            | 505                      | 47.4%           |
| Sequential                            | 658                      | 32.7%           |
| Batching $m = 5$                      | 613                      | 37.2%           |
| Threshold $K = 2$ (OPTN)              | 621                      | 36.6%           |
| Threshold $K = 5$                     | 581                      | 40.4%           |
| Threshold continuous ( $K = \infty$ ) | 508                      | 47.2%           |

Better mechanisms produce shorter waits *and* higher transplant rates. Threshold continuous matches omniscient on both. The collapse under sequential at strong lemons is severe: only 33% of patients ever receive an organ within the simulation horizon, with an average wait of 658 kidneys for those who do.

This is the strongest aggregate efficiency comparison in the study. Mechanisms that prevent kidney discards translate directly into shorter patient waits and higher transplant rates.

## 6.5 Are Any Patients Worse Off Under Threshold Mechanisms?

The aggregate numbers above could mask a real concern: maybe threshold mechanisms reduce discards but at the cost of taking kidneys away from priority-1 patients who would have received them under batching or sequential. To check, we run a **common random numbers** comparison: same patient population, same kidney stream, same random decision draws, but four different mechanisms. We track each individual patient’s outcome under each mechanism.

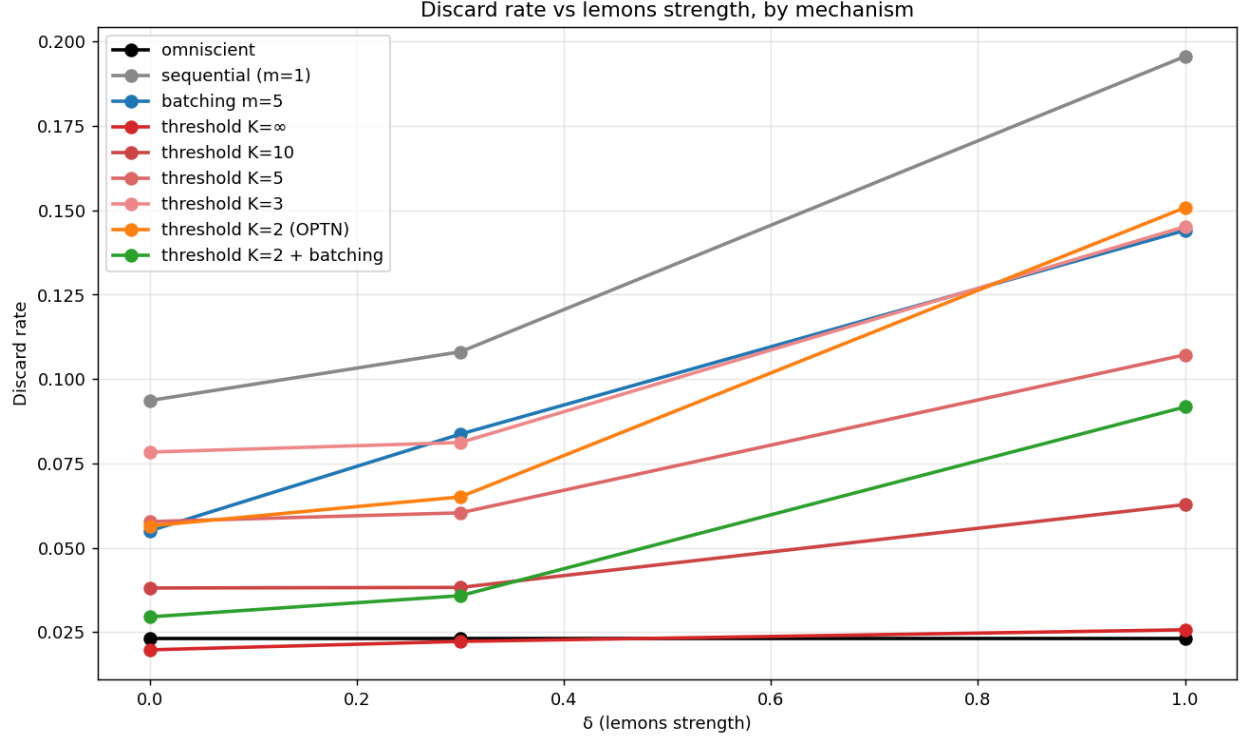


Figure 2: **Discard rate as a function of lemons strength  $\delta$ , by mechanism.** Threshold continuous (dark red, bottom) and omniscient (black, dotted, partially hidden behind threshold continuous) are essentially flat around 2%. As  $\delta$  rises, sequential climbs sharply (reaching nearly 20% at  $\delta = 1.0$ ), with batching ( $m = 5$ , blue) increasing more slowly. The coarse threshold mechanisms ( $K = 10$ ,  $K = 5$ ,  $K = 3$ ,  $K = 2$ ) form an intermediate band; their slopes get steeper as  $K$  decreases. The hybrid threshold  $K = 2 +$  batching (green) outperforms both pure batching and pure  $K = 2$  across the entire range. The figure visually demonstrates that threshold continuous neutralizes the lemons effect, while sequential and batching are exposed to it.

This gives us a true patient-level Pareto comparison: under mechanism A but not under mechanism B, how many patients are there? And vice versa?

The results in Figure 3 are striking. Summing across all 8 replications:

**At  $\delta = 0$ :** Threshold continuous Pareto-dominates batching (341 patients better off under threshold continuous, 0 worse off). Threshold continuous also Pareto-dominates sequential (899 better, 0 worse). The same is not true for threshold  $K=2$  OPTN versus batching at  $\delta = 0$ —batching helps 82 patients that threshold  $K=2$  doesn't, and threshold  $K=2$  helps 2 that batching doesn't.

**At  $\delta = 0.3$ :** Threshold  $K=2$  (OPTN) strictly Pareto-dominates batching (238 better off, 0 worse off). Threshold continuous strictly dominates batching (748 better, 0 worse). Threshold  $K=2$  also dominates sequential (559 better, 0 worse).

**At  $\delta = 1.0$ :** Threshold continuous strictly dominates batching (1410 better off, 0 worse off) and sequential (2026 better, 0 worse). Threshold  $K=2$  and batching are non-comparable:  $K=2$  helps 86 patients, batching helps 21—neither dominates.

**The clearest finding here is that threshold continuous strictly Pareto-dominates batching, sequential, and the coarse threshold mechanisms at every  $\delta$  we tested.** Under

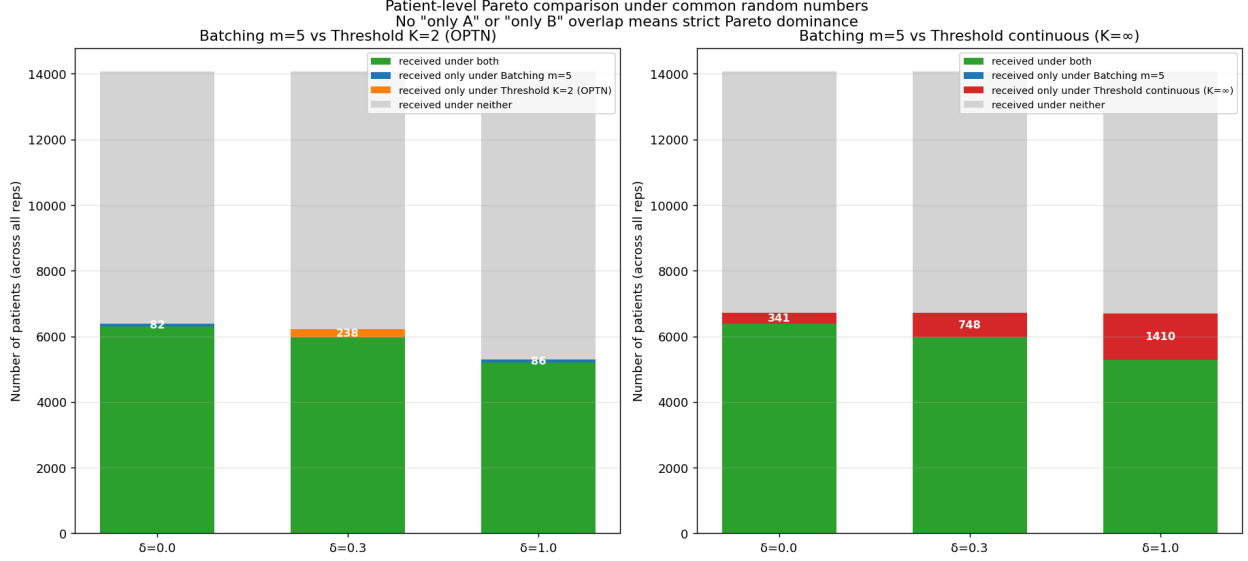


Figure 3: **Patient-level Pareto comparison under common random numbers, summing across 8 replications.** Each stacked bar shows the breakdown of patient outcomes for one pair of mechanisms at one  $\delta$  level: green = received under both, blue/orange/red = received under only one, gray = received under neither. The colored “only” bars are what matter for Pareto comparison—if both “only” bars are nonzero, the mechanisms are non-comparable; if one is zero, the other mechanism strictly Pareto-dominates. **Left panel (batching vs. K=2 OPTN):** at  $\delta = 0$ , batching helps 82 patients more, K=2 helps 2 more (essentially non-comparable, batching slight edge). At  $\delta = 0.3$ , K=2 helps 238 more, batching helps 0—strict dominance reverses. At  $\delta = 1.0$ , the comparison is essentially even (86 vs. 21). **Right panel (batching vs. threshold continuous):** at every  $\delta$ , threshold continuous helps strictly more patients (341, 748, 1410 from left to right) with zero patients worse off. This is strict Pareto dominance at every  $\delta$ .

common random numbers, every single patient who is transplanted under any other mechanism is also transplanted under threshold continuous. The only exception is a small handful of patients under threshold K=2 (OPTN) vs. batching at low  $\delta$ , where the comparison is genuinely non-monotone.

**No patient is worse off under threshold mechanisms compared to batching at  $\delta > 0$ .** The shift in mean placement is not coming at the expense of any individual patient—it’s coming from kidneys that would otherwise have been *discarded* now being placed deep in the queue. The kidneys that were going to the front of the queue under batching still go to the front of the queue under threshold.

## 7 How Do the Mechanisms Compare and Why

The simulations produce a clear ordering at moderate-to-strong observational learning ( $\delta \geq 0.3$ ):

Threshold continuous  $\succ K=10 \succ K=5 \succ$  Batching  $\gtrsim K=2$  (OPTN)  $\succ$  Sequential.

At  $\delta = 0$  (no observational learning), the ordering is less stark: batching, threshold continuous, and the various coarse thresholds are all comparable, while sequential lags. The case for fine-grained threshold mechanisms only becomes overwhelming when lemons effects are substantial.

The mechanism behind this ordering is straightforward once stated. **Each decline made by a non-accepting patient costs the system in two ways: it uses up one hour of the kidney’s**

**12-hour budget, and it amplifies the lemons signal for the next patient.** Mechanisms that minimize unnecessary declines win on both counts.

**Sequential** is the worst because every patient it offers to gets a decline-or-accept opportunity, including patients whose thresholds are nowhere near the kidney’s quality. A patient with  $\tilde{\theta}_i = 0.80$  being offered a kidney with  $\theta = 0.10$  will reject—but their rejection costs the kidney an hour and adds to the lemons cascade.

**Batching** improves on sequential by reaching five patients per hour rather than one, and by freezing the lemons signal within batches. But it still wastes offers on threshold-violating patients (now five at a time).

**Threshold mechanisms** improve on both by routing the kidney directly to patients who have pre-declared willingness. They skip non-eligible patients without cost. With perfect (continuous) declarations, the mechanism essentially never wastes an offer on a non-accepter. With coarse declarations, some within-bucket waste remains: patients whose declared bucket *includes* the kidney’s quality but whose precise threshold *doesn’t* still get offered and decline. The coarser the buckets, the more such waste.

A useful way to see this is the **gap to omniscient**, the difference between a mechanism’s discard rate and omniscient’s. Omniscient has no waste at all—every offer goes to someone who accepts. The gap to omniscient measures how much waste a real mechanism produces. At  $\delta = 1.0$ :

| Mechanism                | Gap to omniscient |
|--------------------------|-------------------|
| Threshold continuous     | 0.3 pp            |
| Threshold $K = 10$       | 4.0 pp            |
| Threshold $K = 5$        | 8.4 pp            |
| Batching $m = 5$         | 12.1 pp           |
| Threshold $K = 2$ (OPTN) | 12.8 pp           |
| Sequential               | 17.3 pp           |

This decomposition reveals that the lion’s share of the inefficiency in current allocation is coming from offering to non-acceptors. Once you give patients a way to opt in or out, even moderately finely, you recover most of the omniscient gain.

## 8 Coarsening: How Bucket Granularity Affects Efficiency

A natural question is how coarse a threshold mechanism can be while still capturing most of the gains. We test buckets of  $K = 2, 3, 5, 10, \infty$ , focusing on how coarsening affects discard rate as  $\delta$  varies.

The pattern in Figure 4 is mostly clean: **discard rate is decreasing in  $K$** , with one exception, and **the cost of coarsening grows with  $\delta$** . Going from  $K = \infty$  to  $K = 2$  costs:

- 3.6 pp at  $\delta = 0$
- 4.3 pp at  $\delta = 0.3$
- 12.5 pp at  $\delta = 1.0$

So coarse buckets are more painful under stronger observational learning. The intuition is that under coarse buckets, patients within the eligible set still produce some declines (because their precise threshold may exceed the kidney’s quality even though their bucket includes it). These

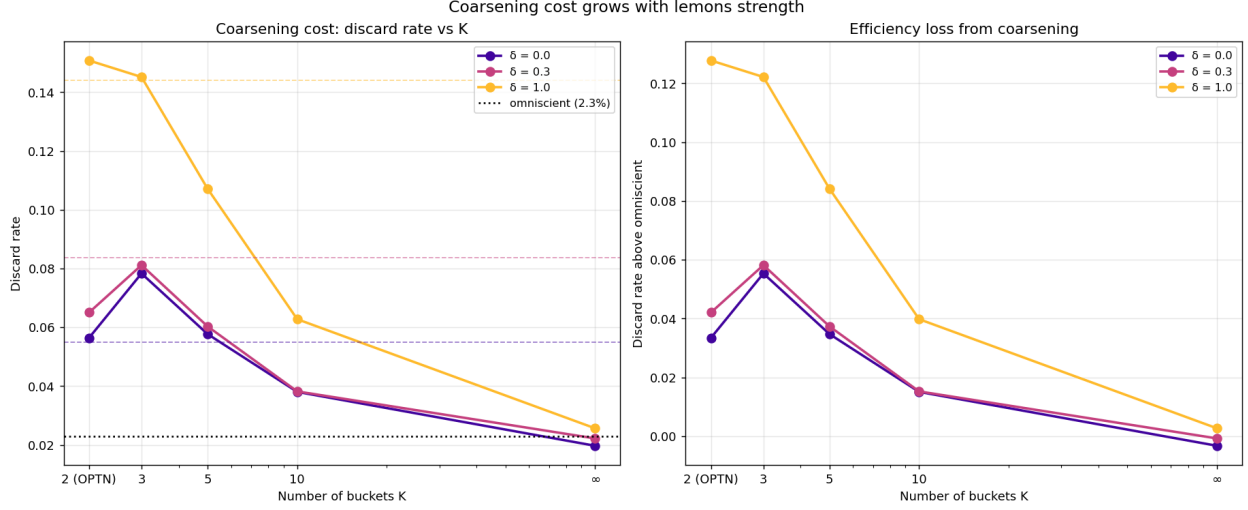


Figure 4: **Coarsening effect on threshold mechanisms.** **Left panel:** discard rate as a function of  $K$  (log-spaced), with one line per  $\delta$  value. Lower is better. The black dotted line is omniscient. Dashed colored lines (mostly off the visible range at high  $\delta$ ) are the batching reference for the corresponding  $\delta$ . **Right panel:** efficiency loss (discard rate above omniscient) as a function of  $K$ . This shows the same data but with the omniscient baseline subtracted, making the coarsening cost easier to read. At  $\delta = 0$  (dark blue), the loss is small at every  $K$ . At  $\delta = 1.0$  (gold), the loss grows substantially as  $K$  shrinks. Note the slight non-monotonicity at  $K = 3$ : with uniform cutoffs at  $\theta = 1/3, 2/3$ , the bucket structure doesn't match the patient threshold distribution, and  $K = 3$  actually underperforms  $K = 2$  (OPTN) which happens to place its cutoff in the heart of the threshold distribution.

within-bucket declines generate lemons signal that hurts subsequent patients. When  $\delta$  is large, those cascading effects compound severely. When  $\delta$  is small, within-bucket declines are merely lost offers, not amplified disasters.

This has a clean policy implication: **if observational learning matters at all in real data, current OPTN policy is leaving substantial efficiency on the table by using only a single binary opt-in for high-KDPI kidneys.** Even moving from  $K = 2$  to  $K = 5$  captures meaningful gains: at  $\delta = 1.0$ ,  $K = 5$  has 10.7% discard vs.  $K = 2$ 's 15.1%.

**The  $K = 3$  anomaly.** There is one apparent anomaly worth flagging.  $K = 3$  underperforms  $K = 2$  (OPTN) in our results at all  $\delta$  levels. Specifically: at  $\delta = 0$ ,  $K = 3$  has 7.8% discard vs.  $K = 2$  OPTN's 5.6%. This is because the  $K = 3$  cutoffs are uniformly placed at  $\theta = 0.33, 0.67$ , which don't align with the patient threshold distribution. Most patients have thresholds below 0.5 (since  $\text{Beta}(2,5)$  is right-skewed with mean 0.29), so  $K = 3$  ends up with very unbalanced bucket populations.  $K = 2$  OPTN, by contrast, happens to place its cutoff at  $\theta = 0.15$ , which is in the heart of the patient threshold distribution and gives a meaningful split.

This raises a finer point: **the placement of cutoffs matters more than the number of buckets.** A well-chosen 2-bucket mechanism beats a poorly-chosen 3-bucket one. The current OPTN cutoff at KDPI = 85 (our  $\theta = 0.15$ ) appears to be near-optimal among binary cuts in our calibrated model. A  $K = 3$  mechanism would need to use cutoffs that adapt to the patient population to outperform  $K = 2$  OPTN.

**Hybrid: threshold + batching.** We also test a  $K = 2$  OPTN mechanism with batching within eligible patients ("threshold-batched  $K = 2$ "). This hybrid achieves the best of both worlds:

it uses self-selection to filter ineligible patients, and uses batching to freeze the lemons signal within the eligible set. At  $\delta = 1.0$ , this mechanism achieves 9.2% discard, beating both pure  $K = 2$  (15.1%) and pure batching (14.4%)—coming closer to threshold continuous’s 2.6%. Batching and threshold mechanisms appear to be **complements, not substitutes**.

## 9 Marginal Kidney Allocation

The original motivation for this study was the marginal kidney problem. About 45% of high-KDPI kidneys ( $\text{KDPI} \geq 85$ , our  $\theta \leq 0.15$ ) are discarded in current U.S. practice (Mohan et al., 2018), compared to under 5% of premium kidneys. The aggregate discard rate is dominated by these marginal kidneys—they are where the action is.

Figure 5 makes this concrete: at  $\delta = 1.0$ , the population-wide discard rate for batching is 14.6%, but its discard rate restricted to marginal kidneys is 75.8%. Threshold  $K = 2$  (OPTN) goes from 14.6% all-kidney discard to 40.1% marginal-kidney discard. Threshold continuous, by contrast, has 2.2% all-kidney discard and 13.7% marginal-kidney discard—essentially matching omniscient on both. **The mechanisms differ massively on marginal kidneys and only modestly on premium kidneys.**

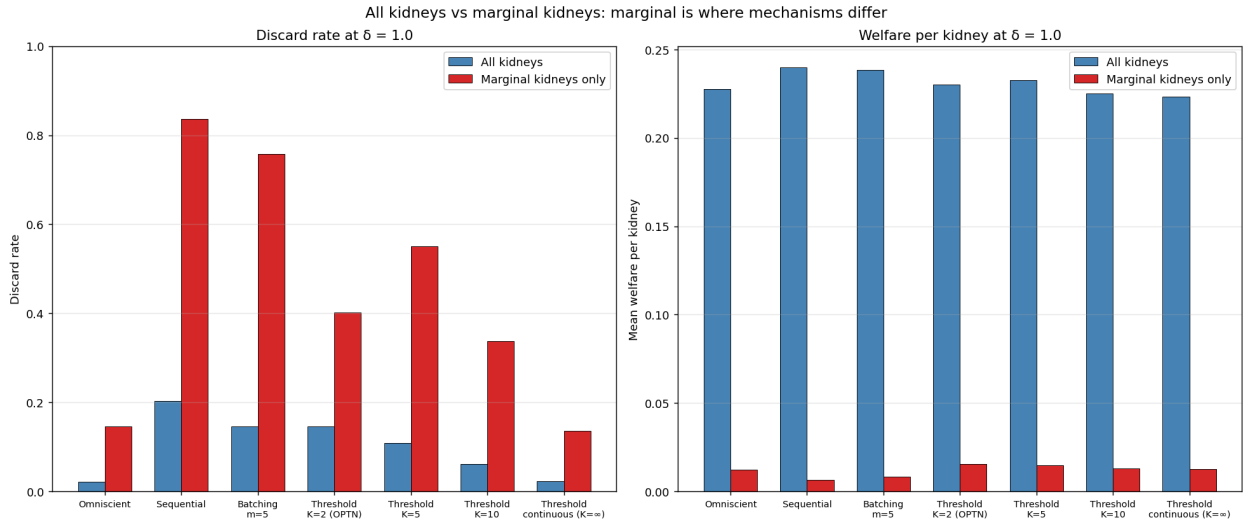


Figure 5: **All kidneys vs. marginal kidneys, comparing discard rate and welfare across mechanisms, at  $\delta = 1.0$ .** The blue bars (“All kidneys”) use every kidney that arrived during the test window; the red bars (“Marginal kidneys only”) restrict to  $\theta \leq 0.15$  ( $\text{KDPI} \geq 85$ ). **Left panel:** the marginal-kidney red bars are dramatically taller than the all-kidney blue bars under sequential and batching, but the two bars are roughly equal under threshold continuous. This shows that the mechanism’s failure mode is concentrated on marginal kidneys. **Right panel:** welfare per kidney. The all-kidney welfare is dominated by premium kidneys (because  $\theta - \hat{\theta}_i$  is large for them), which is why the blue bars are similar across mechanisms. The marginal-kidney welfare (red bars) shows the mechanism differences clearly: threshold mechanisms produce 2–3 $\times$  the marginal-kidney welfare of sequential or batching.

We restrict our analysis to marginal kidneys (the ones with  $\theta \leq 0.15$ , comprising 15% of arrivals) and ask three questions: How often does each mechanism allocate marginal kidneys? Where do those marginal kidneys go in the queue? And how well-matched are the recipients?

## 9.1 Marginal Kidney Discard Rate and Welfare

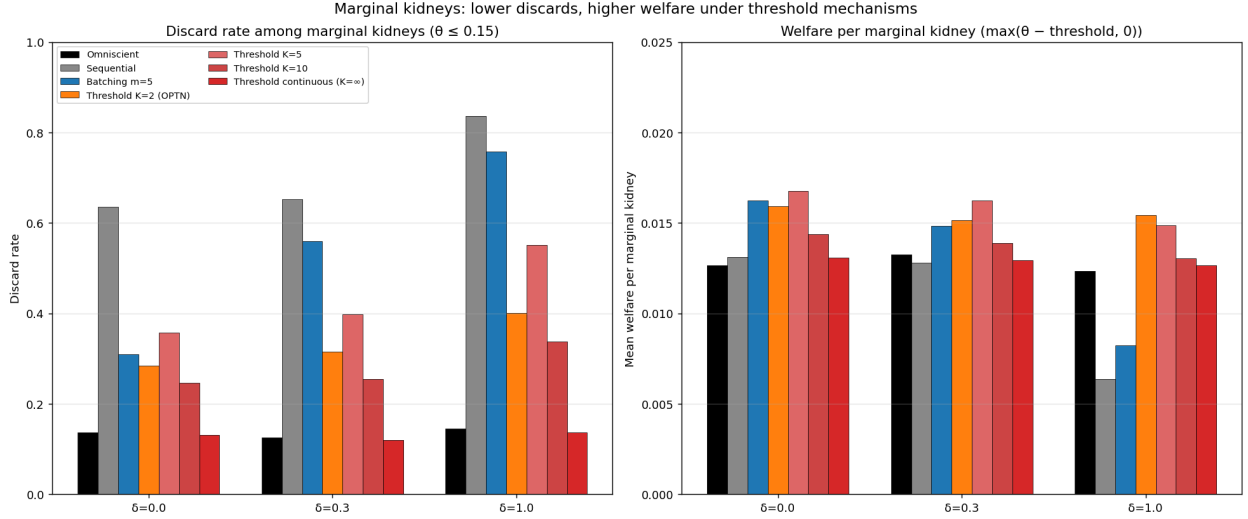


Figure 6: **Marginal kidney discard rate and welfare per marginal kidney by mechanism and  $\delta$ .** Marginal kidneys are those with  $\theta \leq 0.15$  (KDPI  $\geq 85$ ), comprising 15% of arriving kidneys. **Left panel:** discard rate among marginal kidneys. Sequential discards 64–84% of marginals across  $\delta$ , batching 31–76%, threshold  $K = 2$  (OPTN) 28–40%. Threshold continuous and omniscient both sit around 13–15%—the floor where no further allocation is possible because no remaining patient in the queue has a low enough threshold. **Right panel:** welfare per marginal kidney, computed as  $\max(\theta - \tilde{\theta}_i, 0)$  if allocated, else 0. Threshold mechanisms produce 2–3 times higher welfare than sequential or batching. Interestingly, the coarse thresholds ( $K = 2$ ,  $K = 5$ ) slightly outperform the continuous threshold on this metric—they allocate fewer marginal kidneys but with more surplus per allocation.

The pattern at the population level is amplified for marginal kidneys (Figure 6). At  $\delta = 1.0$ , the marginal-kidney discard rates are:

| Mechanism                | Marginal-kidney discard |
|--------------------------|-------------------------|
| Omniscient               | 14.5%                   |
| Threshold continuous     | 13.7%                   |
| Threshold $K = 10$       | 33.7%                   |
| Threshold $K = 2$ (OPTN) | 40.1%                   |
| Threshold $K = 5$        | 55.1%                   |
| Batching $m = 5$         | 75.8%                   |
| Sequential               | 83.6%                   |

Compare this to the population-wide discard rate at  $\delta = 1.0$ : batching 14.4%, threshold  $K = 2$  15.1%, threshold continuous 2.6%. The mechanisms differ much more on marginal kidneys than on the population as a whole. **Marginal kidneys are where the mechanism matters.**

Omniscient itself discards about 14% of marginal kidneys—these are kidneys for which no one in the queue has a low enough threshold to accept. They’re genuinely unallocatable given the patient population. The threshold continuous mechanism essentially achieves this floor. Coarser thresh-



old mechanisms suffer from within-bucket cascades. Batching and sequential fail catastrophically because they spend their 12-hour budget offering to high-threshold patients who decline.

**An interesting non-monotonicity in marginal-kidney discard.**  $K = 2$  (OPTN) outperforms  $K = 5$  on marginal kidneys (40.1% vs. 55.1% at  $\delta = 1.0$ ), even though  $K = 5$  beats  $K = 2$  on aggregate discards (10.7% vs. 15.1%). The reason is that the  $K = 2$  cutoff at  $\theta = 0.15$  exactly defines the marginal kidney boundary, so the marginal-kidney bucket gets only patients who genuinely want marginal organs. The  $K = 5$  uniform cutoffs put marginal kidneys in a bucket  $[0, 0.2)$  that includes some patients with thresholds in  $(0.15, 0.20)$  who would not accept the most marginal organs—producing within-bucket cascades. **The current OPTN policy, by accident or design, has placed its cutoff exactly where it does the most good for marginal kidneys.**

## 9.2 Where Do Marginal Kidneys Go?

We define a recipient’s **match quality** as  $\theta - \tilde{\theta}_i$ : the gap between the kidney’s quality and the recipient’s reservation threshold. A larger gap means the kidney exceeds the patient’s minimum by more—they are getting a kidney they more clearly wanted.

For marginal kidneys (low  $\theta$ ), recipients must have low thresholds. Where in the queue are those low-threshold patients?

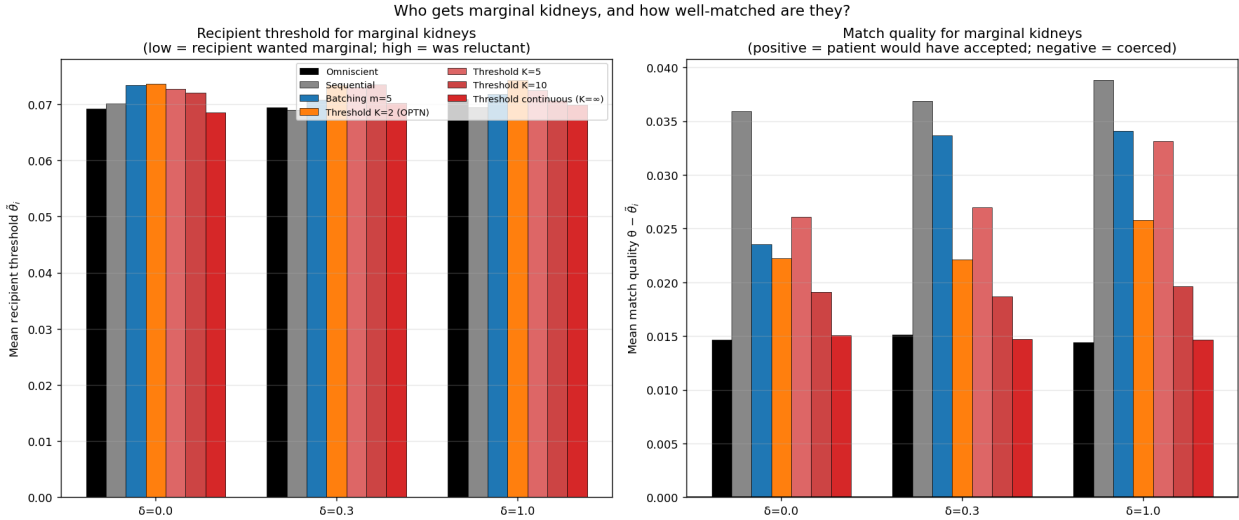


Figure 7: **Recipient threshold (left) and match quality (right) for marginal kidneys.** The recipient threshold is the patient’s true threshold  $\tilde{\theta}_i$ ; the match quality is  $\theta - \tilde{\theta}_i$ . In every mechanism, marginal kidneys go to patients with very low thresholds (around 0.07—well below the population mean of 0.29). Match quality is positive everywhere, meaning every recipient genuinely wanted their kidney. Sequential and batching produce higher match quality (0.034–0.039) than threshold mechanisms (0.014–0.026), because they only manage to place marginal kidneys with patients who have substantially lower thresholds—everyone in between declined. Threshold mechanisms allocate to a wider range of low-threshold patients, including those whose threshold is just below the kidney quality.

At all  $\delta$  levels (Figure 7), the mean recipient threshold for marginal kidneys is approximately  $\tilde{\theta}_i = 0.07$  across all four mechanisms. This is much lower than the population average ( $\tilde{\theta}_i \approx 0.29$ ). So in every mechanism, marginal kidneys go to patients with very low reservation thresholds—patients who explicitly want them.



The mean match quality at  $\delta = 1.0$ :

| Mechanism                | Mean match quality (marginal kidneys) |
|--------------------------|---------------------------------------|
| Sequential               | +0.039                                |
| Batching $m = 5$         | +0.034                                |
| Threshold $K = 2$ (OPTN) | +0.026                                |
| Threshold continuous     | +0.015                                |
| Omniscient               | +0.014                                |

Match quality is positive in every mechanism. **No patient receives a kidney they didn’t want**—the threshold gate in the behavioral model ensures this. But mechanisms differ in *how strictly* they match. Sequential and batching only manage to place marginal kidneys with patients who have substantially lower thresholds (because everyone in between declined). Threshold mechanisms more precisely route marginal kidneys to patients whose threshold is just below the kidney quality.

### 9.3 Allocation Depth for Marginal Kidneys

How deep into the queue does each mechanism reach for marginal kidneys? The answer matters for the fairness story.

Figure 8, from the common-random-numbers analysis, breaks allocation position by kidney quality. For premium kidneys ( $\theta > 0.7$ ), all mechanisms allocate at position 1–2: every patient near the front of the queue accepts a premium kidney, so the mechanism barely reaches into the queue. For marginal kidneys ( $\theta < 0.15$ ), the mechanisms diverge.

Under batching, marginal kidneys go to position 10–25 on average. Under threshold continuous, they go to position 100+ on average (the log-scale axis on the bottom row makes this readable). The reason: threshold continuous searches past every patient whose declared threshold rules out this kidney. It walks all the way to the rare patient who explicitly opted in.

**This is the placement-tradeoff finding restated.** Threshold mechanisms achieve their low marginal-kidney discards by reaching very deep into the queue—much deeper than sequential or batching. But this depth is *exclusively* on marginal kidneys. For middle-quality and premium kidneys, threshold mechanisms allocate at the front of the queue just like everyone else.

Combined with the Pareto finding from §6.5, this means: threshold continuous reaches deep for marginal kidneys, but those deep-position recipients are not “stealing” from front-of-queue patients—the front-of-queue patients are getting their (better) kidneys under threshold continuous just as they do under batching. The deep reaches are pulling out kidneys that would otherwise have been discarded.

### 9.4 Coarsening Effect on Marginal Kidneys

The coarsening analysis from §8 can be repeated specifically on marginal kidneys, and the picture sharpens substantially.

Figure 9 shows the marginal-kidney-restricted coarsening analysis. Three observations:

**The  $K=2$  OPTN cutoff is exceptionally well-placed for marginal kidneys.** The cut-off at  $\theta = 0.15$  exactly defines the marginal kidney boundary. Patients in bucket 0 are exactly those who genuinely want marginal kidneys. This is why  $K = 2$  outperforms  $K = 5$  on marginal kidneys despite  $K = 5$  having finer overall granularity. Under  $K = 5$ , the marginal kidney bucket  $[0, 0.2)$  includes some patients with thresholds in  $(0.15, 0.20)$  who decline the most marginal kidneys, generating within-bucket cascades.

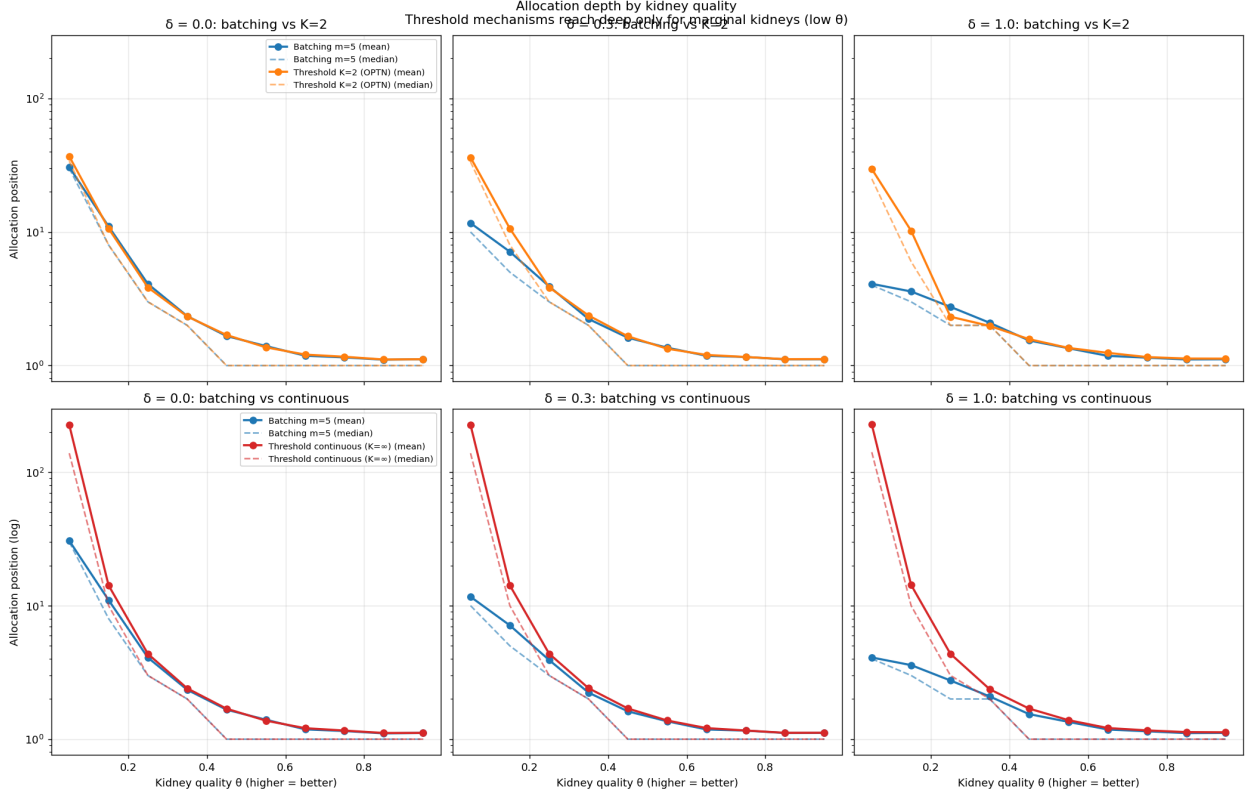


Figure 8: **Allocation depth as a function of kidney quality, by mechanism and  $\delta$ .** Both rows show batching (blue) vs. a threshold mechanism on the same axes. Top row: batching vs. threshold  $K = 2$  (OPTN). Bottom row: batching vs. threshold continuous (note the log  $y$ -axis). Solid lines are means; dashed lines are medians. For premium kidneys ( $\theta > 0.7$ ), all mechanisms allocate at position 1–2: every patient near the front accepts a premium kidney. For marginal kidneys ( $\theta < 0.2$ ), mechanisms diverge. Batching reaches positions 5–25 for marginal kidneys; threshold continuous reaches positions 100+. The “deep reach” of threshold mechanisms is concentrated entirely on marginal kidneys, not premium ones.

**Welfare per marginal kidney is highest at coarse  $K$ .** The middle panel shows  $K = 2$  and  $K = 5$  producing slightly higher welfare than  $K = \infty$  or omniscient. This is because coarse mechanisms allocate to fewer patients but match each one more tightly. (See §9.4 for discussion.)

**Threshold continuous reaches deeply for marginal kidneys.** The right panel shows mean allocation position rising to 135+ at  $K = \infty$ . This is the depth needed to find someone with a low enough threshold. We emphasize again that this deep reach happens only for marginal kidneys—premium kidneys go to the front under every mechanism.

## 9.5 Welfare per Marginal Kidney

Combining discard rate and match quality, we can compute a **welfare metric**:  $\max(\theta - \tilde{\theta}_i, 0)$  if the kidney is allocated, else 0. This gives a per-kidney welfare. Averaging over marginal kidneys at  $\delta = 1.0$ :

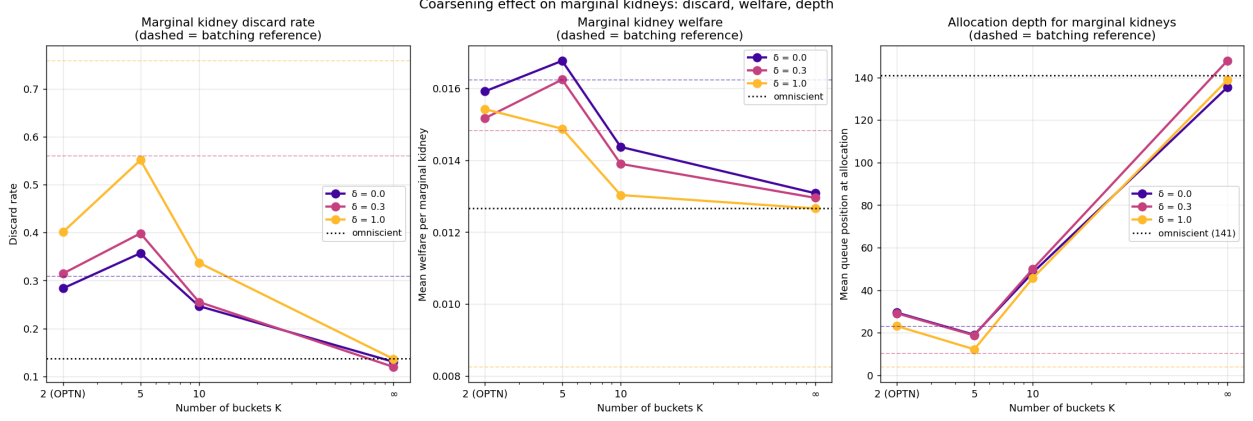


Figure 9: **Coarsening effect on marginal kidneys: discard rate (left), welfare per kidney (middle), and mean allocation position (right) as functions of bucket count  $K$  (log-spaced).** Solid lines: threshold mechanisms at  $\delta = 0$  (purple), 0.3 (pink), 1.0 (gold). Dashed lines: batching reference at corresponding  $\delta$ . Dotted black: omniscient. **Left:**  $K = 2$  (OPTN) achieves lower marginal-kidney discard than  $K = 5$  at every  $\delta$ , because its cutoff at  $\theta = 0.15$  aligns exactly with the marginal kidney boundary.  $K = 5$ ’s uniform cutoffs put marginal kidneys in a bucket that includes too many non-accepting patients, triggering within-bucket cascades. **Middle:** welfare per marginal kidney peaks at  $K = 2$  and  $K = 5$ , slightly above  $K = \infty$  and omniscient. Coarse buckets allocate fewer marginal kidneys but with higher per-kidney surplus. **Right:** mean allocation position rises sharply with  $K$ , but only for marginal kidneys.  $K = \infty$  reaches positions 130–150 on average to find an accepter for the most marginal kidneys.

| Mechanism                | Welfare per marginal kidney |
|--------------------------|-----------------------------|
| Threshold $K = 2$ (OPTN) | 0.015                       |
| Threshold $K = 5$        | 0.015                       |
| Threshold $K = 10$       | 0.013                       |
| Threshold continuous     | 0.013                       |
| Omniscient               | 0.012                       |
| Batching $m = 5$         | 0.008                       |
| Sequential               | 0.006                       |

The threshold mechanisms produce roughly 2x the welfare per marginal kidney that batching does, and 2–3x what sequential does. Interestingly, the coarse threshold mechanisms ( $K = 2$  and  $K = 5$ ) produce slightly higher welfare than threshold continuous and omniscient. The reason: threshold continuous allocates a higher fraction of marginal kidneys (87% vs.  $K = 5$ ’s 45%), but it also reaches very low-threshold patients (mean  $\tilde{\theta}_i = 0.07$  matches the kidney quality more closely, leaving less “surplus” per match). Coarse mechanisms allocate fewer marginal kidneys but with slightly higher per-kidney surplus.

This is a real but subtle finding. **Omniscient maximizes allocation rate; coarse threshold mechanisms maximize per-kidney match surplus.** Which is “better” depends on the policy maker’s objective. If the goal is to reduce discards (public-health framing), omniscient and threshold continuous win. If the goal is to maximize the satisfaction of each transplant recipient, coarse mechanisms might be slightly preferable. In practice, the differences here are small and probably swamped by other considerations.

## 10 Additional Analyses

A few analyses we ran beyond the headline results.

**Common-random-numbers fairness.** We extensively used CRN to verify that the mechanism-level differences we report are not driven by random variation. The CRN comparisons showed strict Pareto dominance of threshold continuous over all other mechanisms at  $\delta \geq 0.3$ , a much stronger claim than just “lower average discard rate.”

**Patient-level transplant rates by initial priority tercile.** Patients in the top tercile by initial priority have a transplant rate of 100% under all mechanisms (they all get an organ eventually). Patients in the middle tercile show large differences: at  $\delta = 1.0$ , threshold continuous gives mid-tercile patients a 41% transplant rate, while batching gives them 0%—they all wait forever. Patients in the bottom tercile (latecomers) are reached only by threshold continuous, and even then only at a 1–2% rate within our simulation horizon.

**Optimal  $K = 2$  cutoff.** We swept the binary cutoff over  $\theta \in \{0.15, 0.20, 0.30, 0.40, 0.50\}$  and found that the cutoff at  $\theta = 0.15$  (the OPTN policy) is at the minimum of the discard-rate curve across all  $\delta$ . Moving the cutoff up reduces eligible-patient pool size and increases discards. The current OPTN binary opt-in is near-optimal as a binary mechanism, which is a nontrivial validation of existing policy.

**Hybrid threshold + batching.** Combining  $K = 2$  OPTN with batching  $m = 5$  (offering simultaneously to 5 eligible patients per hour) achieves 9.2% discard at  $\delta = 1.0$ —between  $K = 2$  alone (15.1%) and threshold continuous (2.6%). Batching adds value on top of self-selection because within-bucket declines still trigger cascades that batching can interrupt.

**Distribution of allocation depth.** Cumulative distribution plots of allocation positions show that 50% of all allocations happen at position 1 under every mechanism. The mechanisms differ in the *tail*—threshold continuous has a long tail reaching position 100+, while batching is bounded at position 60. This tail is entirely populated by marginal kidneys, as the by-quality plot shows.

## 11 Limitations and What Real Implementation Would Need

Our simulation makes a number of simplifying assumptions worth flagging before concluding.

**Patient thresholds are exogenous and fixed.** Real patients adjust their preferences over time (sicker, more desperate, more accepting), and would adjust strategically in response to mechanism changes. The Su-Zenios incentive-compatibility result for threshold mechanisms assumes patients know their own reservation thresholds—a non-trivial assumption that the structural literature (Agarwal et al., 2025) addresses with detailed choice modeling.

**Single-quality kidney type.** We collapsed all medically relevant features into one scalar  $\theta$ . Real kidneys have many dimensions: HLA matching, donor age, hepatitis C status, cold ischemic time. Real OPTN opt-ins are multi-attribute (high-KDPI, DCD, HCV+, increased-risk donor). Extending threshold mechanisms to multi-attribute would require either many separate binary opt-ins (manageable) or a multi-dimensional declaration (complex, may not be incentive-compatible).

**Priority is purely time-on-list.** We abstract from medical urgency, geographic priority, pediatric status, and HLA matching. Real OPTN priority is multi-factor. Some of these factors would correlate with thresholds in ways our model doesn’t capture (very sick patients have lower thresholds and also higher priority, which would correlate the priority and threshold dimensions).

**The lemons effect is assumed uniform across kidney quality.** In reality, observational learning may operate more strongly for marginal kidneys (where “why did they decline?” has more semantic content) than for premium kidneys. A quality-varying  $\delta$  would likely amplify the threshold

mechanism’s advantage on marginal kidneys.

**The 10% baseline decline is calibrated, not empirical.** The logistic baseline  $\alpha = \text{logit}(0.9)$  is chosen to be plausible but not directly fit to data. Real decline rates among opted-in patients vary by center, kidney type, and donor characteristic. Sensitivity to this parameter is moderate—pushing the baseline to 80% or 95% changes magnitudes but not qualitative orderings.

**Continuous time and instantaneous decisions.** Real OPTN offers are made on a more complex temporal structure involving simultaneous parallel offering, electronic acceptance, transplant team scheduling, and so on. We treat each hour as a clean decision point, which is a simplification.

Within these limitations, the qualitative findings are robust. The strict Pareto dominance of threshold continuous (under common random numbers, at  $\delta > 0$ ) does not depend on calibration choices—it holds across the parameter ranges we tested.

## 12 Conclusion

This study compares four mechanisms—omniscient, sequential, batching, and threshold—for allocating kidneys with viability deadlines, under a unified behavioral model where patients have private thresholds and prior declines update beliefs about kidney quality. The findings:

1. **Threshold continuous (continuous self-declaration of reservation thresholds) effectively eliminates the kidney discard problem** caused by observational learning. It achieves omniscient-level discard rates regardless of lemons strength.
2. **Under common random numbers, threshold continuous strictly Pareto-dominates batching, sequential, and the coarse threshold mechanisms at  $\delta > 0$ .** No patient is worse off; many are better off. The mean-placement difference is not coming at the expense of any front-of-queue patient.
3. **Coarse threshold mechanisms (current OPTN binary opt-in) capture much of the gain but not all.** At  $\delta = 1.0$ ,  $K = 2$  OPTN achieves 15% discard rate vs. threshold continuous’s 3%. Going to  $K = 5$  or  $K = 10$  recovers most of the remaining gain.
4. **Marginal kidneys are where mechanisms differ most.** Mechanism choice has a 70-percentage-point effect on marginal-kidney discard rate at  $\delta = 1.0$ , vs. a 17-percentage-point effect on the population-wide rate.
5. **Threshold mechanisms route marginal kidneys deeper into the queue, but only marginal kidneys.** Premium kidneys go to the front under every mechanism. The “deep reach” of threshold continuous is pulling out kidneys that would otherwise be discarded, not redistributing kidneys that would have gone to high-priority patients.
6. **Batching and threshold mechanisms are complements.** A hybrid threshold + batching mechanism outperforms either alone.

The current OPTN system uses a binary opt-in for high-KDPI kidneys, which corresponds to our  $K = 2$  mechanism. Our analysis suggests this is the right kind of mechanism but coarser than ideal. Moving to finer-grained opt-ins ( $K = 5$  or  $K = 10$ ) would capture substantial additional efficiency, especially if observational learning is as strong in practice as Zhang’s empirical work suggests.

Whether these findings hold up in real allocation depends crucially on whether the lemons interpretation of decline cascades is correct. If declines are independent (as Agarwal et al. (2021)

argue), threshold mechanisms still help, but the gap to batching shrinks substantially. Discriminating between these two views requires panel data on offer responses with rich kidney covariates, and remains the most important empirical question raised by this work.

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